José Ignacio Maldonado Pérez

Computer Systems Engineer — Process Automation & API Integration

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PROFESSIONAL SUMMARY

Computer Systems Engineer with 3+ years of experience in operations, testing and server diagnostics. I design and run end-to-end automations: the n8n workflow, the OAuth integration with third-party APIs and the deployment on Google Cloud with retries, idempotency and duplicate control. Two of my personal projects run unattended in production today.

TECHNICAL SKILLS

Automation	n8n, webhooks, scheduled jobs, pipeline orchestration, retries, idempotency
API integration	REST, OAuth 2.0, YouTube Data API, Instagram Graph, TikTok Content Posting API, Facebook Graph
Applied AI	Google Gemini API, Google Cloud Text-to-Speech, prompt design, output validation
Backend	Node.js, Express 5, Firebase Cloud Functions (Node.js 22), Python (basic), SQL
Cloud & infra	Google Cloud Run, Cloud Storage, Firebase, Docker, Linux, Git/GitHub
Operations & quality	SLAs, incident management, Power BI, functional testing, technical documentation

SELECTED PROJECTS

CIENCIA COOL

Personal project · 2026

Content automation: from news item to published video, with no manual step · n8n · Node.js · Google Cloud · FFmpeg · Demo / Code

- Fully automated flow: pulls news items from RSS, writes the script with Gemini, synthesises the narration, selects stock footage and renders the video to MP4 with FFmpeg.
- Orchestrated in n8n on Docker: scheduled trigger, duplicate control by job id, retries with an adjusted timeout and automatic deletion once publishing finishes.
- OAuth 2.0 integrations with YouTube, Instagram, TikTok and Facebook, with automatic token refresh and publishing to all four networks in a single run.
- Built to fail safely: API key, duplicate lock (409/429), idempotency and Cloud Run size limits. About 3 minutes per video and cents per year to run.

NEXO BOT

Personal project · 2026

Automated WhatsApp support: menu, booking and owner alerts · WhatsApp Cloud API · Gemini · Firestore · Cloud Run · Demo

- Support bot on Meta's official API: deterministic menu, appointment booking against real availability and an automatic owner notification when a booking comes in.
- HMAC-SHA256 signed webhook, swappable storage adapter and Docker deployment on Cloud Run; the same engine serves WhatsApp, the web demo and the CLI.
- Resilience: Gemini model chain with a timeout and a local FAQ fallback, so the critical booking path never depends on the model.

WORKLISH

Personal project · 2026

Serverless backend for an AI English tutor · Cloud Functions · Firestore · Gemini · Demo

- Eleven functions that verify the Firebase token, apply the plan quota and call Gemini to correct text, score pronunciation and keep the conversation going.
- Automatic subscription sync from RevenueCat to Firestore and security rules that scope every document to its owner.

PROFESSIONAL EXPERIENCE

FOXCONN

Jun 2025 – present

Jr. Test Engineer · Electronics manufacturing, NVIDIA project

- Run tests on nodes and servers and diagnose hardware and software failures.
- Support the NVIDIA account: reproduce defects, document findings and track reported cases.

UNIVERSIDAD DE DESARROLLO NACIONAL (UDN)

May 2024 – Sep 2025

Lecturer, Engineering Department

- Taught engineering courses and developed course content and teaching materials.

MEGACABLE

Sep 2022 – Dec 2023

Jr. NOC Operations Engineer · Networks and telecommunications

- Handled service incidents, classified them by severity and followed up in line with SLAs.
- Built and analysed Power BI reports to identify operational improvement opportunities.

EDUCATION, CERTIFICATIONS & LANGUAGES

Education	B.Eng. in Computer Systems Engineering, UDN, 2018–2021. Federal professional licence 13727204.
Certifications	EF SET English Certificate B1 · Power BI Career Accelerator · Ask Questions to Make Data-Driven Decisions (Coursera).
Languages	Spanish (native) · English B1, EF SET certified.